

# NP Project Proposal

## “Hybrid TCP/UDP Chat System”

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### 1. Title

GUI-Based Intelligent Multi-Client Chat System Using TCP/UDP Hybrid Protocol

### 2. Objective

To develop a real-time chat system with GUI that ensures reliable messaging using TCP and fast updates using UDP.

### 3. Problem Statement

Traditional communication systems rely either on TCP for reliability or UDP for speed. This project combines both protocols to achieve efficient communication.

### 4. Methodology

TCP will be used for message delivery and file transfer, while UDP will handle real-time features such as typing indicators and online presence.

### 5. Features

- Multi-client chat
- GUI-based interface
- Typing indicator (UDP)
- Online user detection
- Message broadcasting

### 6. Tools & Technologies

C#, .NET, Socket Programming, WinForms, VS Code

## **7. Expected Outcome**

A fully functional hybrid communication system demonstrating networking concepts.

## **8. Future Work**

Encryption, file transfer, mobile app integration, cloud deployment.